Why Your Vector Search Is Silently Wrong (And How to Fix It)

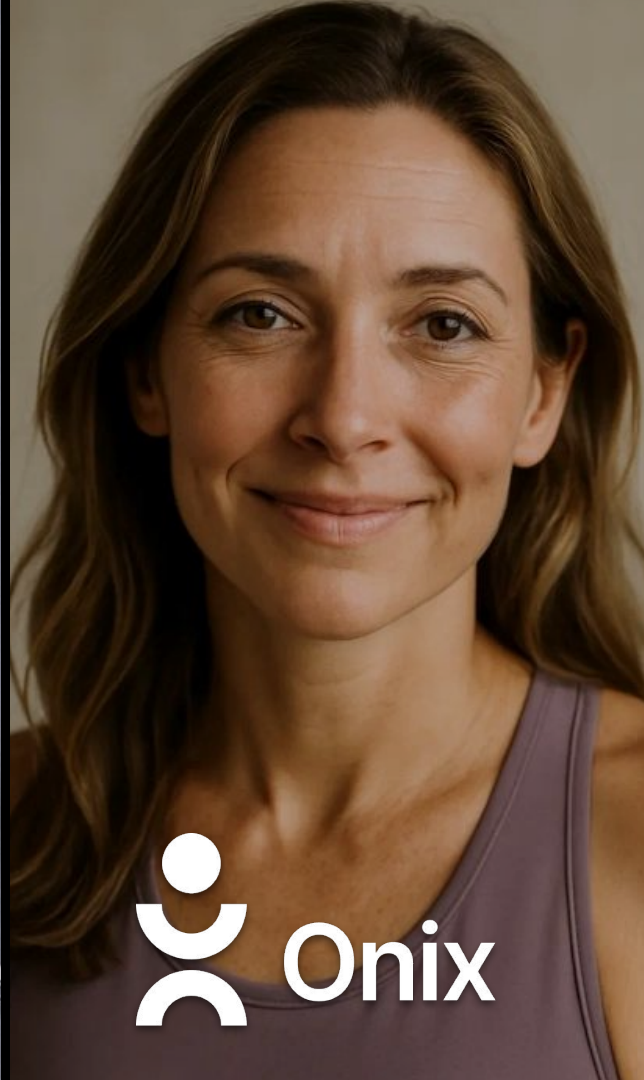
Nicholas Nadeau
Co-Founder & CTO, Onix

Onix

```python
np.linalg.norm(embedding)
# Expected: 1.0
# Actual: 23.7
```

Onix

AON3D


Onix

One of the world's most recognized consulting firms.

Broken RAG system.

Seven-figure annotation project proposed.

The vectors weren't normalized.

Onix

```
embedding = embedding / np.linalg.norm(embedding)
```

Onix

What Are Embeddings, Actually?

Onix

Embedding Model

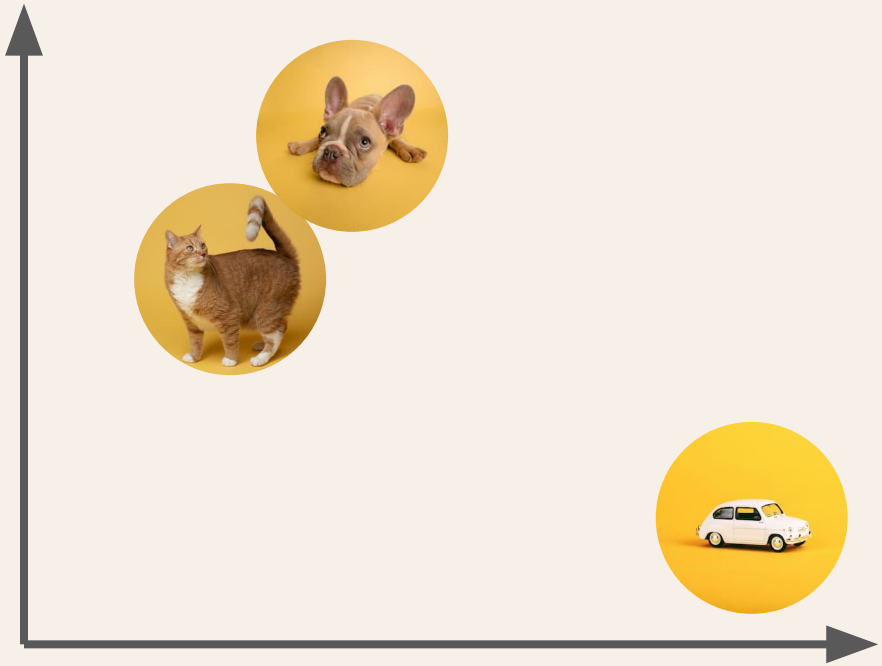
[0.2, -0.5, ..., 0.3, 0.9]

Onix

Embeddings Intuition

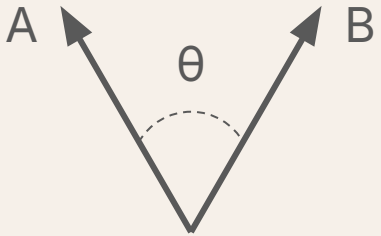
Embedding models turn inputs (e.g., text, images, sound) into coordinates.

Similar things end up close together.



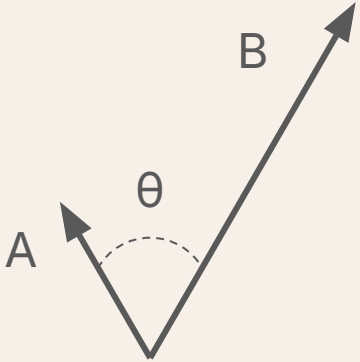
Onix

How We Measure "Close"



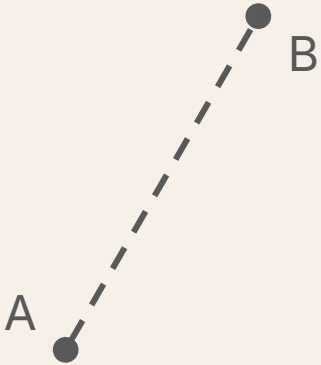
Cosine Similarity

The angle between two arrows



Dot Product

Angle AND length



Euclidean Distance (L2)

Straight-line distance between points

Onix

Many systems optimize cosine similarity by assuming your vectors are pre-normalized and running dot product.

If your vectors aren't normalized, that assumption breaks silently.

Onix

Cosine similarity, fully expanded:

$$\cos(\theta) = (a \cdot b) / (\|a\| \times \|b\|)$$

Onix

What the database actually runs (the optimization):

```
dot_product(a, b) = a · b
# Assumes ||a|| = ||b|| = 1, so the denominator disappears
```

Onix

What happens when your norm is 23.7 instead of 1.0:

```
# What you asked for:
cos(θ) = (a · b) / (23.7 × 18.4)  →  0.82  ✓

# What the database computed:
dot_product(a, b)                 →  357.6  ✗

# The ranking is now dominated by vector magnitude, not direction
# Longer vectors win regardless of semantic similarity
```

Onix

This Isn't Just a Query-time Problem

- The distance metric shapes your index at build time.
- **Hierarchical Navigable Small Worlds (HNSW)**: top-performing, graph-based algorithm for Approximate Nearest Neighbor (ANN) search in high-dimensional spaces. Metric determines which vectors become neighbors.
- **Inverted File with Flat Compression (IVFFlat)**: an ANN search index that partitions data into k clusters (centroids) using K-means. Metric drives the cluster assignments.
- You can't change the metric after the index exists.

Onix

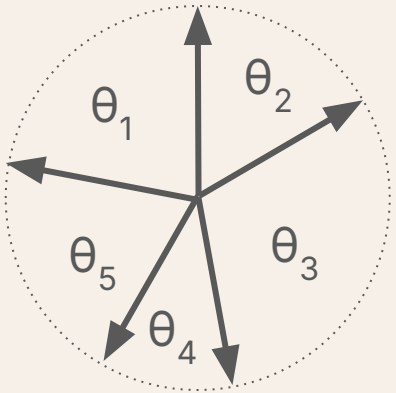
What metric is your database using?

Database	Default
pgvector (Supabase)	Depends on query operator
Pinecone	Cosine
Weaviate	Cosine (runs dot product internally)
Qdrant	You must choose
Chroma	Euclidean distance (L2)
FAISS	Euclidean distance (L2)

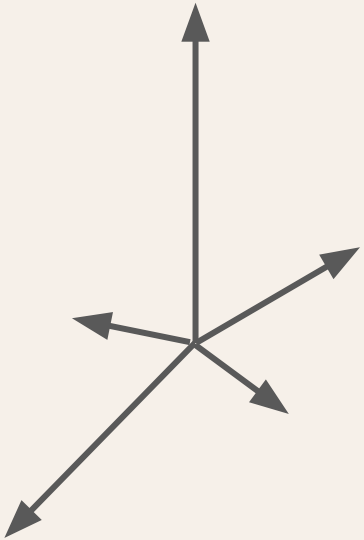
Most developers never check.

Onix

Normalized vs. Unnormalized



Measuring Meaning



Measuring Magnitude

Onix

But My Provider Normalizes For Me

I tested 25+ embedding models last week. Most do.

Model	Provider	Norm
baai/bge-m3	OpenRouter	1.00
google/gemini-embedding-001	OpenRouter	1.00
openai/text-embedding-3-large	OpenRouter	1.00
openai/text-embedding-ada-002	OpenRouter	1.00
qwen/qwen3-embedding-8b	OpenRouter	1.00
BAAI/bge-base-en-v1.5	Hugging Face	1.00
Qwen/Qwen3-Embedding-8B	Hugging Face	1.00

Onix

Some Don't: Close enough to fool you

Model	Provider	Norm
mistralai/codestral-embed-2505	OpenRouter	0.987941
Qwen/Qwen3-Embedding-0.6B	Hugging Face	0.998936
qwen/qwen3-embedding-4b	OpenRouter	0.999953
mistralai/mistral-embed-2312	OpenRouter	1.00002
nvidia/llama-embed-nemotron-8b	Hugging Face	1.001647

Onix

Some Don't: Your search is returning garbage

Model	Provider	Norm
Supabase/gte-small	Hugging Face	8.8125
ibm-granite/granite-embedding-small-english-r2	Hugging Face	17.27116
ibm-granite/granite-embedding-english-r2	Hugging Face	30.422783

Onix

Same Vendor. New Model. Broken Geometry.

Model	Provider	Norm
ibm-granite/granite-embedding-107m-multilingual	Hugging Face	1.00
ibm-granite/granite-embedding-125m-english	Hugging Face	1.00
ibm-granite/granite-embedding-278m-multilingual	Hugging Face	1.00
ibm-granite/granite-embedding-30m-english	Hugging Face	1.00
ibm-granite/granite-embedding-small-english-r2	Hugging Face	17.27116
ibm-granite/granite-embedding-english-r2	Hugging Face	30.4227

Onix

From Enterprise Bug to Everyone's Problem

Onix

2023 : ML engineers building RAG pipelines. Months of development. Custom infrastructure.

2026 : Any developer with an API key. Twenty minutes to ship. Vibe coded.

The easier it is to build, the less you understand what you've built.

I'm not anti-vibe-coding.

I'm anti-vibe-debugging.

Onix

The Diagnostic Toolkit

Onix

```python
import numpy as np

def check_embedding_health(embeddings):
    norms = np.linalg.norm(embeddings, axis=1)
    print(f"Min norm:  {norms.min():.4f}")
    print(f"Max norm:  {norms.max():.4f}")
```

Onix

```python
def normalize(embeddings):
    norms = np.linalg.norm(embeddings, axis=1, keepdims=True)
    return embeddings / norms
```

Two lines. That's the fix.

Onix

```python
def safe_embed(text, model):
    embedding = model.encode(text)
    norm = np.linalg.norm(embedding)
    if abs(norm - 1.0) > 0.01:
        raise ValueError("Embedding not normalized")
    return embedding
```

Don't trust. Verify.

Onix

```python
# Before quantization
norm_before = np.linalg.norm(embedding)  # 1.0

# After int8 quantization + dequantization
quantized = (embedding * 127).astype(np.int8)
recovered = quantized.astype(np.float32) / 127
norm_after = np.linalg.norm(recovered)   # 0.97-1.03
```

You normalized. Your database un-normalized.

Onix

The Diagnostic Pipeline

1. Embed your text
2. Check the norm. Is it ~1.0?
3. If no: normalize
4. Store in your database
5. Retrieve and check norm again (quantization drift?)
6. If drifted: re-normalize after retrieval
7. Use cosine similarity unless you've verified normalization
8. Match your query operator to your index operator class

Onix

pgvector Operators

Operator	Metric	Index Operator Class
< = >	Cosine distance	vector_cosine_ops
< # >	Negative inner product	vector_ip_ops
< - >	Euclidean distance (L2)	vector_l2_ops

Onix

The Silent pgvector Trap

```sql
-- You created the index with cosine
CREATE INDEX ON items
  USING hnsw (embedding vector_cosine_ops);

-- But you query with inner product
SELECT * FROM items
  ORDER BY embedding <#> query_embedding
  LIMIT 5;

-- pgvector silently ignores your index.
-- Full sequential scan. No error. No warning.
```

Onix

The enterprise team spent months planning an annotation project and a RAG rebuild.

The fix took fifteen minutes.

Onix

Before you add complexity, verify your foundations work.

Onix

```python
np.linalg.norm(embedding)
# Expected: 1.0
# Actual: 23.7
```

Onix

Nicholas Nadeau
Co-Founder & CTO, Onix

Onix